

untitled2

November 20, 2025

```
[1]: !pip install -q langchain langchain-google-genai google-generativeai  
     ↪python-dotenv
```

42.0/42.0 kB

2.1 MB/s eta 0:00:00

```
[2]: import os  
  
# Replace this with your real Gemini api key  
os.environ["GOOGLE_API_KEY"] = "AIzaSyCnUtX11UqnN2d89I52fdcDE0d-eN5bbxk"  
  
# If you prefer not to hardcode it, you can do:  
# from getpass import getpass  
# os.environ["GOOGLE_API_KEY"] = getpass("Enter your Gemini API key: ")
```

```
[7]: from langchain_google_genai import ChatGoogleGenerativeAI  
     from langchain_core.prompts import ChatPromptTemplate  
     from langchain_core.output_parsers import StrOutputParser  
  
# 1. LLM: Gemini model via LangChain  
llm = ChatGoogleGenerativeAI(  
    model="gemini-2.5-flash", # fast & free tier friendly  
    temperature=0.3  
)  
  
# 2. System + user prompt template  
system_prompt = """  
You are a helpful AI assistant for a Security Operations Center (SOC).  
Your tasks:  
- Analyze the user's question,  
- Explain security / SIEM / SOC concepts clearly,  
- Give step-by-step, beginner-friendly answers,  
- When relevant, connect ideas to SIEM rules, alerts, and incident response.  
  
If the question is not about security, still answer politely and helpfully.  
Keep answers concise but clear.  
"""
```

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human_prompt = "{user_input}"

prompt = ChatPromptTemplate.from_messages(
    [
        ("system", system_prompt),
        ("human", human_prompt),
    ]
)

# 3. Chain = Prompt -> Gemini -> Plain text
chain = prompt | llm | StrOutputParser()

```

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[ ]: def run_agent():
    print("Security Assistant Agent (LangChain + Gemini)")
    print("Type 'exit' or 'quit' to stop.\n")

    while True:
        user_input = input("You: ")
        if user_input.lower() in ["exit", "quit", "q"]:
            print("Agent: Goodbye! ")
            break

        # Invoke the LangChain chain
        response = chain.invoke({"user_input": user_input})
        print("\nAgent:", response, "\n")

# Run the agent
run_agent()

```

Security Assistant Agent (LangChain + Gemini)
Type 'exit' or 'quit' to stop.

You: Hello

Agent: Hello! Welcome to the SOC.

How can I help you today? Do you have any questions about security, SIEM, incident response, or anything else I can assist with?

You: What is STEM EXPLAIN

Agent: STEM is an acronym that stands for:

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*   **S**cience
*   **T**echnology
*   **E**ngineering

```

* **M**athematics

It refers to a broad range of academic disciplines and professional fields that are crucial for innovation, research, and development in many areas, including computing, healthcare, space exploration, and more.

While STEM itself isn't a security concept, many roles within a Security Operations Center (SOC) heavily rely on STEM skills, particularly in areas like:

* **Technology:** Understanding networks, operating systems, and security tools.

* **Engineering:** Designing and implementing secure systems and solutions.

* **Mathematics:** For cryptography, data analysis, and understanding algorithms.

* **Science:** Applying logical and analytical thinking to investigate incidents and understand attacker methodologies.

You: Who are you

Agent: Hello! I am an AI assistant designed to help you in the Security Operations Center (SOC).

My purpose is to assist with:

* **Analyzing security questions.**

* **Explaining security, SIEM, and SOC concepts.**

* **Providing step-by-step guidance.**

* **Connecting ideas to SIEM rules, alerts, and incident response.**

Think of me as a helpful resource to make your security tasks clearer and more manageable. How can I assist you today?

Task 2 – AI Agent (LangChain + Gemini)

Internship Assessment – AI/ML (AI Agents Domain)

Company: Goklyn Pvt. Ltd.

Objective

The goal of Task 2 is to create a simple AI agent using LangChain that can analyze user input and generate meaningful responses. I used Google Gemini (free API) as the underlying large language model.

Tools & Technologies

Python

Google Colab

LangChain

Gemini 1.5 Flash model

langchain-google-genai package

Approach

1. Installed necessary libraries in Google Colab.

2. Configured Gemini API key using environment variables.

3. Created a LangChain agent using:

ChatGoogleGenerativeAI (Gemini model)

ChatPromptTemplate containing system instructions

StrOutputParser for clean text output

4. Implemented an interactive chat loop so the agent can:

Receive user input

Analyze the query

Generate clear, structured responses

The agent was designed to give security/SIEM-focused explanations, but it can also answer general technical questions.

Behavior

Understands natural language input

Analyzes user questions

Generates helpful, beginner-friendly answers

Responds especially well to security, SOC, and SIEM-related topics

Example tests included:

"What is a SIEM system?"

"Explain brute-force login attacks."

"How can AI help threat detection?"

The agent responded accurately and consistently.

Result

A fully working AI assistant was created using LangChain + Gemini that meets all requirements:

Uses LangChain

Uses a real LLM (Gemini)

Analyzes input

Produces intelligent responses

Runs interactively

This completes Task 2 as required in the assessment.